Can the Strongest Practical Mitigations Omit LoRA? A Frozen-Encoder Follow-Up on BRACS and TCGA-UT

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Abstract

Three mitigations of class imbalance were among the most effective when a Virchow2 encoder was adapted through low-rank adaptation (LoRA): train-time logit adjustment, post-hoc logit adjustment, and the curriculum of data augmentation (CUDA). At an imbalance ratio of 100, they recovered between about two-fifths and nearly all of the lost balanced accuracy. This experiment asks whether the three methods lose at most one percentage point of balanced accuracy at that ratio when the encoder stays frozen. Six paired contrasts, one per dataset and method, decide the question; omitting LoRA counts as supported only if all six show no material loss. Only the two BRACS logit-adjustment contrasts show no material loss. BRACS CUDA and the two TCGA-UT logit-adjustment contrasts are inconclusive, and TCGA-UT CUDA shows a LoRA advantage of more than one point. LoRA can therefore be omitted for logit adjustment on BRACS, but not across the tested scope. Without LoRA, logit adjustment gains about twice as much over cross-entropy as with it, whereas CUDA gains about the same, so each CUDA contrast largely reflects the gap between the two cross-entropy controls.

1 Motivation

The mitigation studies trained a low-rank adaptation (LoRA) update of Virchow2 (Zimmermann et al., 2024) together with the classification head (Queisler, 2026a,c). Logit adjustment (LA) recovered the most accuracy on TCGA-UT; the curriculum of data augmentation (CUDA) (Ahn et al., 2023) led in the other settings (Queisler, 2026b). LoRA makes every fit train part of the encoder. If the strongest methods keep their accuracy on a frozen encoder, that step can be dropped. This follow-up asks: *do these methods retain their balanced accuracy when the foundation model is not adapted at all?*

Each method is compared with its own LoRA counterpart under the same imbalanced allocation. A plain linear probe is a reference, not a substitute for applying the same mitigation without LoRA.

2 Design

Both datasets use the 30 split-draw shards of the mitigation studies (three patient-disjoint splits, ten draws each) at the severe imbalance $\rho = 100$. The adapted representation reuses the final matching records of the BRACS and TCGA-UT mitigation studies. The frozen representation trains only a linear head on pretrained Virchow2, with the same training patients, patches, order, labels, class counts, and validation and test partitions.

Training follows the first stage of the source studies: AdamW, batch size 256, learning rate 10^{-4}, weight decay 10^{-4}, a cosine schedule, and 600 (BRACS) or 2,400 (TCGA-UT) updates, with empirical row sampling. Cross-entropy (CE) and train-time LA train on cached encoder features. Post-hoc LA applies each strength to the stored CE head. CUDA keeps its image

operations, per-class learning levels, classifier checks, and constants ($\gamma = 0.6$, check size 10, maximum strength 10); augmented images pass through the frozen encoder, and only the head trains. The adapted runs did not record their head initialization, so data and inference are paired but initialization is not.

Validation patient-macro balanced accuracy alone selects the strength per method, representation, split, and draw from the grids in Table 1. No strength is transferred from the adapted runs, and no grid is extended after test results are visible.

Method	BRACS	TCGA-UT
Train-time LA, τ	0.5, 1.0	0.25, 0.5, 1.0
Post-hoc LA, τ	0.5, 1.0, 1.5, 2.0	0.25, 0.5, 1.0, 1.5, 2.0
CUDA, p_{aug}	0.25, 0.5, 1.0	0.25, 0.5, 1.0
Updates	600	2,400

Table 1: Fixed strength grids at $\rho = 100$. CE has no strength; it is the control and the source head of post-hoc LA. CE: cross-entropy; LA: logit adjustment; CUDA: curriculum of data augmentation.

3 Evaluation

For each dataset and method, the primary contrast at $\rho = 100$ is

$$\delta = \text{BA}(\text{without LoRA}) - \text{BA}(\text{with LoRA}), \tag{1}$$

in percentage points, with patient-macro balanced accuracy (BA) pooled over splits and draws. As in the mitigation studies, uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws within each split. Both representations share the replicate weights, so every contrast is paired. Because six contrasts decide one question, each interval is a two-sided percentile interval from the $0.05/12$ and $1 - 0.05/12$ quantiles (about 99.17%), which gives the six together Bonferroni 95% familywise coverage. Such extreme quantiles need more replicates than the 95% intervals of the mitigation studies: this experiment uses 10,000 instead of 2,000, leaving about 42 rather than 8 replicates beyond each bound.

A loss counts as material if it exceeds one point. A contrast shows *no material loss* if its lower bound lies strictly above -1: omitting LoRA then costs at most one point. It shows a *LoRA advantage* if its upper bound lies strictly below -1. Any other outcome, including a bound equal to -1, is *inconclusive*. Omitting LoRA is supported only if all six contrasts show no material loss. That result would mean "no material accuracy loss from omitting LoRA in the tested scope", not equality or the absence of any LoRA benefit. A nonsignificant difference alone does not show that a loss is not material.

Secondary diagnostics are descriptive: absolute BA, each representation's gain over its CE control, tail recall, and validation selections.

4 Results

Table 2 reports the six primary contrasts at $\rho = 100$. Only two of the six contrasts show no material loss, so omitting LoRA is not supported across the tested scope; each contrast still stands as an individual outcome. The CE controls reach a balanced accuracy of 31.17 (frozen) and 31.57 (adapted) on BRACS, and 67.32 (frozen) and 69.95 (adapted) on TCGA-UT.

The CE controls explain much of this pattern. Frozen minus adapted CE is -0.40 points $[-1.45, 0.56]$ on BRACS but -2.64 points $[-3.12, -2.16]$ on TCGA-UT, so on TCGA-UT LoRA

Dataset	Method	Without LoRA	With LoRA	δ (pp)	99.17% interval	Decision
BRACS	Train-time LA	35.75	33.54	2.22	[0.43, 3.97]	no material loss
BRACS	Post-hoc LA	35.34	33.51	1.83	[0.11, 3.46]	no material loss
BRACS	CUDA	33.01	34.17	-1.16	[-2.80, 0.40]	inconclusive
TCGA-UT	Train-time LA	70.84	71.49	-0.65	[-1.21, -0.11]	inconclusive
TCGA-UT	Post-hoc LA	70.83	71.37	-0.54	[-1.14, 0.06]	inconclusive
TCGA-UT	CUDA	68.20	70.97	-2.77	[-3.38, -2.13]	LoRA advantage

Table 2: Balanced accuracy (BA, %) at $\rho = 100$ without and with LoRA, and their paired difference δ with a 99.17% percentile interval (Bonferroni over six contrasts). A lower bound above -1 shows no material loss; an upper bound below -1 is a LoRA advantage; anything else is inconclusive. LA: logit adjustment; CUDA: curriculum of data augmentation.

already improves the unmitigated model by more than one point. Each contrast δ equals this CE gap plus the difference between the two representations' gains over their CE controls (Table 3). Both forms of LA gain about twice as much without LoRA as with it: 4.58 and 4.17 against 1.96 and 1.94 points on BRACS, and 3.52 and 3.51 against 1.54 and 1.41 on TCGA-UT. On BRACS, this larger gain outweighs the small CE gap, so both LA contrasts show no material loss. On TCGA-UT, it closes most of the 2.64-point gap but leaves both lower bounds below -1. CUDA gains about the same in both representations (1.83 against 2.59 points on BRACS, 0.89 against 1.02 on TCGA-UT), so its contrasts largely inherit the CE gap: the TCGA-UT LoRA advantage of 2.77 points is close to the CE gap of 2.64.

		Gain over CE (pp)		Tail recall (%)	
Dataset	Method	Without LoRA	With LoRA	Without	With
BRACS	CE	—	—	2.2	5.1
BRACS	Train-time LA	4.58 [2.94, 6.39]	1.96 [0.79, 3.19]	23.4	20.2
BRACS	Post-hoc LA	4.17 [2.50, 6.03]	1.94 [0.80, 3.23]	24.2	22.3
BRACS	CUDA	1.83 [0.82, 2.97]	2.59 [1.46, 3.73]	6.0	11.9
TCGA-UT	CE	—	—	53.6	61.2
TCGA-UT	Train-time LA	3.52 [3.13, 3.93]	1.54 [1.21, 1.86]	68.1	69.4
TCGA-UT	Post-hoc LA	3.51 [3.12, 3.91]	1.41 [1.18, 1.66]	68.2	69.9
TCGA-UT	CUDA	0.89 [0.49, 1.32]	1.02 [0.71, 1.33]	63.3	62.9

Table 3: Without LoRA, logit adjustment gains about twice as much over CE and lifts tail recall to near the adapted level; CUDA gains about the same in both representations. Gain in balanced accuracy over the representation's own CE control at $\rho = 100$, with 99.17% paired intervals, and mean patient-macro recall of the third of classes with the smallest allocation (descriptive, no intervals). CE: cross-entropy; LA: logit adjustment; CUDA: curriculum of data augmentation.

Tail recall shows where the LA gains come from. In the third of classes with the smallest allocation, frozen CE recalls 2.2% on BRACS and 53.6% on TCGA-UT, against 5.1% and 61.2% with LoRA. Both forms of LA raise frozen tail recall to 23–24% on BRACS, above their adapted values, and to 68% on TCGA-UT, one to two points below them. The frozen head third pays for this: its recall falls from 57.0% to 39.5–40.8% on BRACS and from 79.6% to 72.5–72.6% on TCGA-UT. Frozen CUDA lifts BRACS tail recall only to 6.0%; its BRACS gain lies in the middle third, whose recall rises from 21.4% to 39.2%. On TCGA-UT, frozen CUDA raises tail recall to 63.3% but lowers head recall to 72.4%, which leaves a net gain below one point.

Validation selects stronger corrections without LoRA. On BRACS, frozen train-time LA selects the full correction $\tau = 1$ in 25 of 30 shards (adapted: 23), post-hoc LA selects it in 21

(adapted: 15), and frozen CUDA selects $p_{\text{aug}} = 1$ in 25 (adapted: 14). On TCGA-UT, both forms of frozen LA select $\tau = 0.5$ in 19 of 30 shards (adapted: 17 and 20). Frozen CUDA selects $p_{\text{aug}} = 0.5$ or 1 in 27 of 30, whereas the adapted runs select 0.25 in 17 and never select 1. Two frozen BRACS selections therefore sit at the upper grid edge in most shards. For CUDA, $p_{\text{aug}} = 1$ is the largest possible probability; for train-time LA, values of τ above 1 remain untested. The grids were fixed before test results and were not extended.

5 Limitations

The follow-up reuses data already inspected in the mitigation studies; it is not an independent test set. It covers three methods, one encoder, one allocation, fixed training recipes, and the existing 30-draw design. Execution devices, precision, and micro-batching differ from the adapted runs, so training is not bitwise comparable. In particular, frozen CUDA augmentation sees bicubic-resized images, matching the cached features, whereas the adapted CUDA runs used a bilinear resize, so CUDA image preprocessing is not paired. Frozen CUDA training forwards also use bf16 autocast. A LoRA advantage under these recipes does not show that every optimizer setting needs LoRA, and a contrast with no material loss does not extend to other methods, encoders, or ratios.

References

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